



GREEN VALLEY MEDICAL CENTER ANALYSIS DASHBOARD

Data Analysis Excel Capstone Project

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Introduction

Green Valley Medical Center is a multi-department healthcare facility that provides a wide range of medical services, including cardiology, orthopedics, neurology, pediatrics, and general medicine.

This Analysis organizes the data into a structured dashboard to help administrators evaluate departmental performance, treatment effectiveness and cost efficiency in order to improve patient recovery outcomes.

The data are routinely collected, it is often stored in raw tables, making for it difficult for hospital administrators and healthcare managers to quickly extract meaningful insights.

Project Objectives

Analyze recovery outcomes by department doctors and treatment type.

Assess and compare Departmental patient outcomes, hospital stay duration, and treatment

efficiency. Explore demographic Impact such as age, gender, and other patient characteristics on

recovery. Assess and compare Departmental patient outcomes, hospital stay duration, and treatment

efficiency. Develop an Interactive Excel dashboard to visualize key metrics and trends.

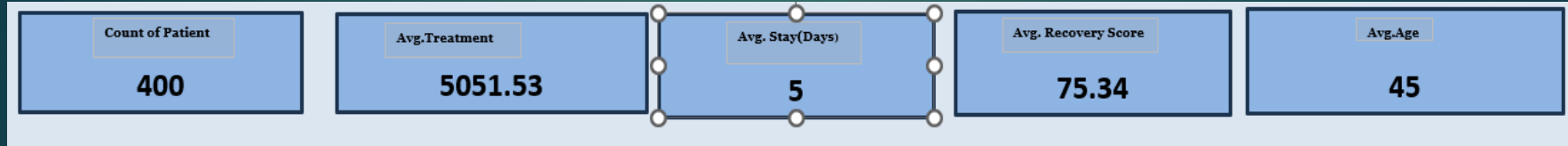


The dataset for this project

This comprises of 400 patient records from Green Valley Medical Center, capturing

information across multiple hospital departments. Examples are Age, Patient ID, gender, departments, treatment type and treatment cost ,Doctor name, hospital Stay and Recovery Score. Age and Treatment cost were categories into the group in additional columns.

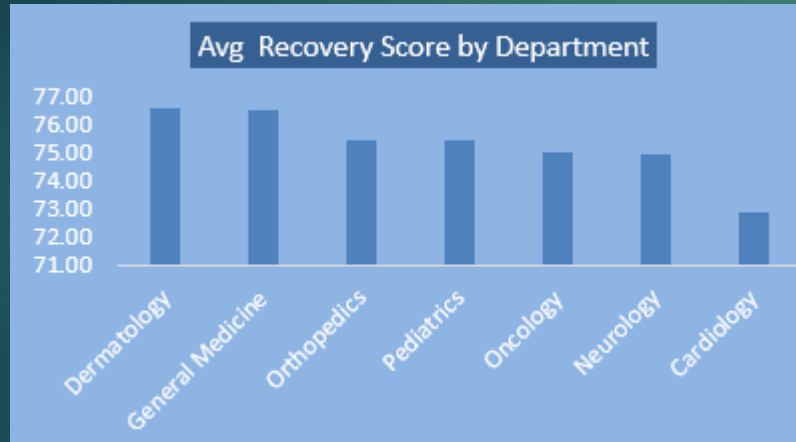
Key Metrics



Insight: The hospital treated 400 of patients with an average recovery score of 75.34 showing positive Conclusions. An Average stay of 5 days and Average treatment cost of 5051.53 suggest moderate resources consumption.

Recommendation: Organization should monitor patients with stays or treatment costs significantly above average to ascertain inefficiencies or complicated cases that may require targeted interventions.

Average recovery score by department



Insight: Dermatology and General Medicine show the highest average recovery scores. While Cardiology ranks lowest.

Recommendation: To improve the average recovery score in lower performing departments. There should be multidisciplinary approach tailored to each specialty's unique challenges.

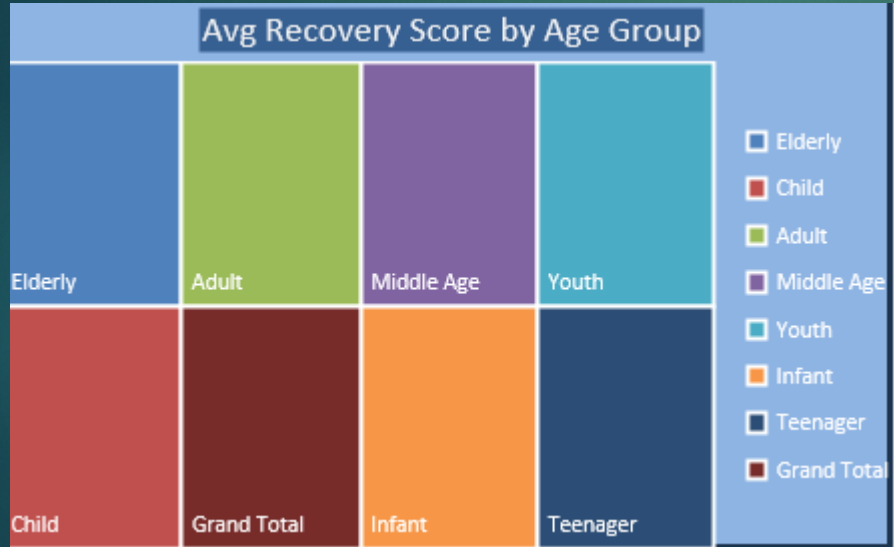
Average recovery score by treatment

Avg Recovery Score by Treatment	
Surgery	76.84
Therapy	76.09
Medication	75.67
Observation	72.86
Grand Total	75.34

Insight: Surgery yielded the highest average recovery score at 76.84 while Therapy follows closely with a score of 76.09

Recommendation: Observation has the lowest average recovery score, conduct a sub analysis to determine if this is due to the nature of the conditions being observed or if those patient would benefit form early introduction of Medication or therapy to bridge the performance

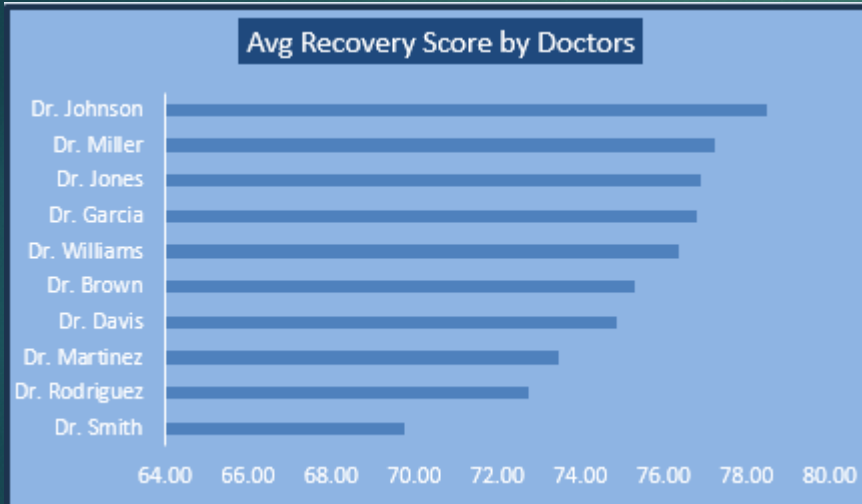
Average Recovery score by Age Category



Insight: Recovery scores show minimal variation across age groups, indicating no clinically significant age-related disparity.

Organization should monitor age specific outcomes to sustain this stability.

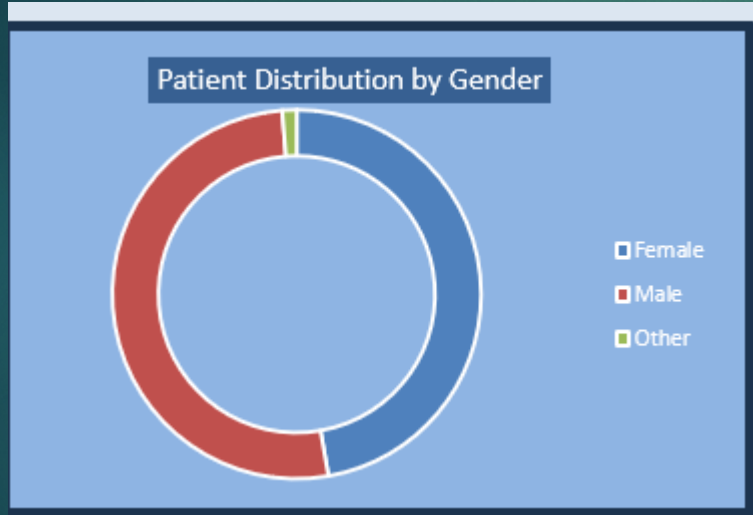
Average recovery score by Doctors



Insight: Dr Johnson has the highest average recovery score while Dr Smith has the lowest average recovery score.

Recommendation: Is to pair top performer like Dr Johnson and Dr miller with those with lower range like Dr Rodriguez and Dr Smith.

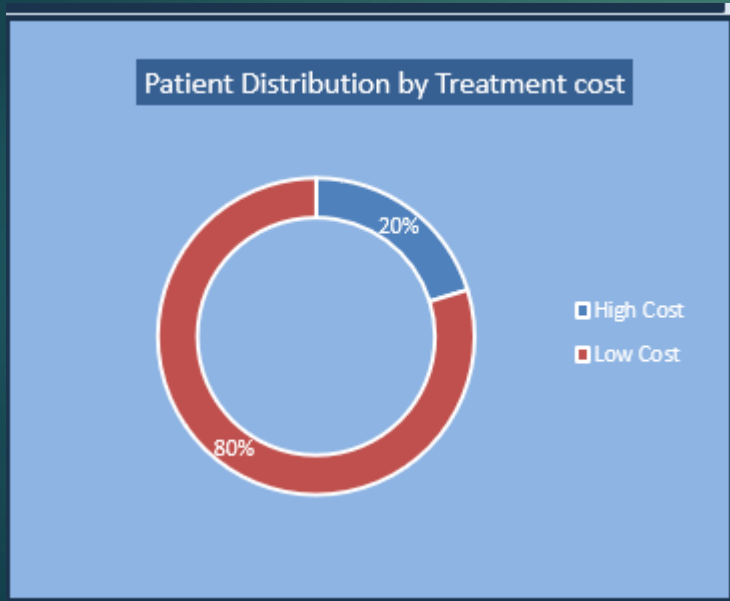
Patient Distribution by Gender



Insight: The distribution shows that the hospital serves a broad diverse population with no significant gender-based skew.

Recommendation: The other category are small. It may be worthwhile to review your intake forms to ensure they are inclusive of all gender identities, ensuring every patient feels accurately represented.

Patient Distribution by Treatment cost category



Insight: The vast majority of the patient population falls into the low-cost treatment category. While a minority group are grouped as high cost.

Recommendation: Recognize rising risk patient who are currently low cost but show clinical markers of becoming high cost.

GREEN VALLEY MEDICAL CENTRE ANALYSIS DASHBOARD

Count of Patient

400

Avg.Treatment

5051.53

Avg. Stay(Days)

5

Avg. Recovery Score

75.34

Avg.Age

45

Observation

Surgery

Therapy

Doctor Name

Dr. Brown

Dr. Davis

Dr. Garcia

Dr. Johnson

Dr. Jones

Dr. Martinez

Dr. Miller

Dr. Rodriguez

Gender

Female

Male

Other

Department

Cardiology

Dermatology

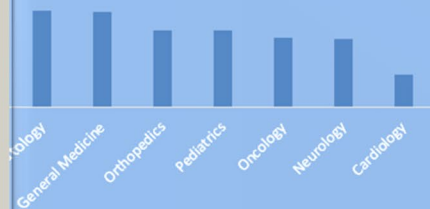
General Medicine

Neurology

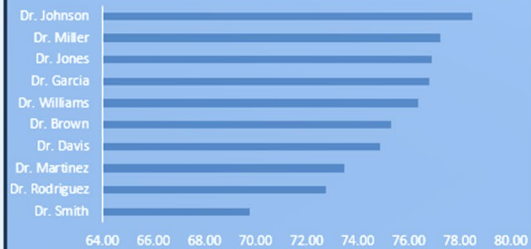
Oncology

Orthopedics

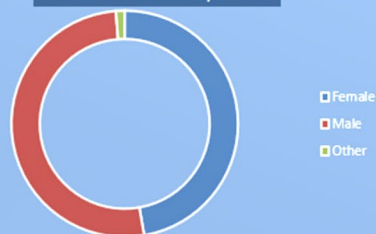
Avg Recovery Score by Department



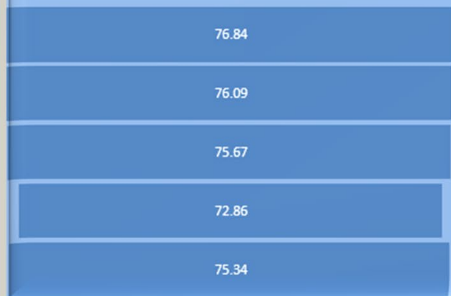
Avg Recovery Score by Doctors



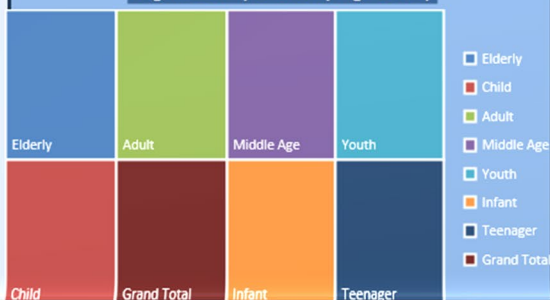
Patient Distribution by Gender



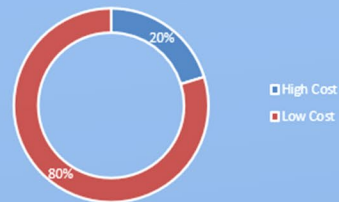
Avg Recovery Score by Treatment



Avg Recovery Score by Age Group



Patient Distribution by Treatment cost





THANKS!